

# YASHWANTH DEVAVARAPU

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## EDUCATION

### KL University

Aug 2021 – May 2025

Bachelor of Technology in Computer Science and Engineering (Hon's); CGPA: 9.6/10

## WORK EXPERIENCE & HACKATHON PARTICIPANT

- Software Engineer - Barclays** Pune, Maharashtra  
*Java 21, Spring Boot, LangGraph, PostgreSQL, AWS* July 2025 – Present
  - Architected an 8-microservice internal developer platform serving 100+ applications, centralizing vulnerability management, FinOps, database operations, and infrastructure automation while automating security scans, CVE detection, Jira ticket creation, and reporting for 50+ engineering teams using mcp tools.
  - Designed a hybrid SES-Keycloak authorization model with dynamic role provisioning, enabling self-service onboarding for 50+ teams.
  - Developed an AWS FinOps module providing cloud cost analytics, anomaly detection, forecasting, and savings recommendations across multiple AWS accounts.
  - Automated GitLab migration and CI/CD onboarding workflows for 100+ repositories, standardizing engineering processes and reducing manual effort.
  - Built security automation platforms covering 15,000+ services and 4,000+ applications, automating container remediation, vulnerability detection, validation pipelines, reducing remediation timelines from days to under one hour.
- Software Engineering Intern - Barclays** Pune, Maharashtra  
*Java, Spring Boot, Apache Ignite, Thymeleaf, Caches, Distributed Systems* June 2024 - July 2024
  - Developed a full-stack cache monitoring platform using Spring Boot, Thymeleaf, and Apache Ignite to track SDS & ESM cache data across distributed systems.
  - Implemented anomaly detection and automated email alerts for cache fluctuations, improving transaction monitoring and system reliability.
- JPMC Hackathon Participant - JPMorgan Chase** Hyderabad, Telangana  
*Java, Spring Boot, SQL, REST APIs, CORS, Backend Development* May 2024
  - Developed a backend platform for NGO operations, implementing CRUD APIs for donor management, beneficiary tracking, and fund distribution while improving SQL query performance by 40%.
  - Integrated frontend and backend services with secure CORS handling, enabling reliable API communication and data flow.

## SKILLS SUMMARY

- Languages & Version Control:** Java, Python, SQL, C++, C, Bash, Linux, Git, GitHub, GitLab
- Frameworks & Libraries:** Spring Boot, FastAPI, Django, LangChain, CrewAI, Hugging Face, REST APIs, WebSockets
- AI & Agent Systems:** LLMs, Generative AI, Prompt Engineering, RAG, Multi-Agent Systems, Agent Orchestration, Transformers, AI Evaluation, Model Benchmarking, Vector Databases
- AI Developer Tools:** GitHub Copilot, Cursor, Claude Code, OpenAI Codex
- Databases & Infrastructure:** MySQL, MongoDB, ChromaDB, FAISS, Apache Ignite, Docker, Kubernetes, GitLab, CI/CD, OpenTelemetry (OTEL), Distributed Systems, YAML, JSON
- Cloud & Deployment:** AWS (Bedrock, SageMaker), LangSmith, LangServe, Microservices, Event-Driven Architecture

## PROJECTS

- ChatQuill – Real-time Chat Application (WebSockets, Backend Development):** Created a real-time chat application with WebSocket-based bidirectional communication, reducing latency by 50%. Implemented STOMP protocol for message broadcasting and user presence tracking. Tech: Spring Boot, Thymeleaf, WebSockets, HTML, CSS
- Daily Expenses Sharing Application (Expense Management, Backend Development):** Constructed a backend system to handle daily expense splitting using equal, exact, and percentage-based methods. Designed REST APIs for user authentication, transaction records, and balance tracking. Tech: Spring Boot, MySQL, JPA, Lombok
- Cure Connect – Healthcare Web App (Web Development, Healthcare):** Established an online healthcare appointment booking system with user authentication and secure patient data management. Integrated MongoDB for data storage and efficient retrieval. Tech: Django, MongoDB, HTML, CSS
- Gen AI Chatbot with SQL Database (LangChain, SQL Integration):** Engineered a chatbot capable of querying SQL databases to fetch and summarize data dynamically, enhancing real-time decision-making. Tech: LangChain SQL Toolkit, PostgreSQL, OpenAI API
- Building a Chatbot that Chats with PDFs (RAG, Document AI):** Enabled AI-powered PDF interaction, allowing users to query documents and receive intelligent responses while maintaining chat history. Tech: LangChain, AstraDB, ChromaDB, OpenAI API

## PUBLICATIONS

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### **Through the Looking Glass: Mirror Reasoning Benchmark for Vision-Language Models** CVPR'26

Developed a benchmark comprising 700+ images and 3,500+ QA pairs to evaluate reflection reasoning capabilities in state-of-the-art vision-language models.

### **EdgeMedBench: Benchmarking Small Language Models for Medical QA** IC4AI'26

First author. Designed and evaluated 45 configurations across five small language models and three quantization levels for medical question answering under edge deployment constraints.

### **ML-Based SDN DDoS Detection: Machine Learning-Enabled IDS** CIMA'26

Designed and evaluated machine learning techniques for software-defined network intrusion and DDoS attack detection.

### **Credit Card Fraud Detection Using Outlier Analysis and Detection** CONIT'24

Proposed anomaly detection and outlier analysis techniques for financial fraud identification.

### **Preprints & Under Review**

- *MVEB: Massive Video Embedding Benchmark* — arXiv'26
- *TRACE: Verifiable RL Environment for Network Troubleshooting Agents* — Submitted to EMNLP'26
- *APISHIFT: Verifiable Agent Learning for API Contract Migration* — Submitted to EMNLP'26
- *CRYO-AGENT: Agentic Cryo-ET Reconstruction with Physics-Guided Retrieval-Augmented Diffusion* — Submitted to ECCV'26
- *TabuLM: Tabular Language Model for Structured Reasoning Tasks* — Submitted to WACV'26

**Research Contributions:** Benchmarking, dataset construction, multimodal AI, vision-language models, video embeddings, agentic AI systems, software engineering agents, medical AI, and verifiable RL environments.

## CERTIFICATIONS & ACHIEVEMENTS

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- **AWS Certified Cloud Practitioner (Certification):** Earned certification validating foundational knowledge of AWS services, cloud concepts, security, pricing, and architecture best practices.
- **Director's Recognition Award:** Awarded for delivering Agentic AI systems that improved automation reliability and reduced user-prone errors across internal workflows.
- **Meta PyTorch OpenEnv Hackathon Finalist:** Ranked among the Top 100 teams nationwide from 31,000+ registrations and 800+ finalist teams at the Meta PyTorch OpenEnv Hackathon.

## ACADEMIC SERVICE

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- **Technical Committee Member – IEM-HEALS 2026:** Selected as a Technical Committee Member for the International Conference on Advancing Science and Technologies in Health Science (IEM-HEALS 2026), supporting research review activities.